

UAV Software Ground School Notes

1. What kind of software & Hardware will be on our drone?

- There's a lot on the drone:
 - WiFi antenna
 - Cube (flight controller)
 - 360 auto-stabilize gimbal cameras
 - GPS receiver (2x) and other sensors
 - NVIDIA Jetson Orin Nano (this is our onboard computer)

A. Full Software/Hardware Spec List

- **Object Detection (Week 3)**
 - It's usually done with OpenCV(trad comp vision library)
 - Now we use, YOLO (manually supervised learning),
 - Our task is labelling data and parameter tuning to fit our model to tasks
- **Aerial Imagery/Stitching (Week 2)**
 - This has been perfected for more than 15 years!
 - Used to use OpenCV but now OpenDroneMap is used (better results)
 - This is done offline

- **Flight Controller (Week 4)**
- There are two parts to an FC:
 - Hardware (Cube Orange+, Pixhawk)
 - Firmware(Stuff like Ardupilot/Betaflight)
- Keeps drone above the ground
- Almost no autonomy, receives commands via MAVLink over a physical link (Ex: UART serial cable) from an onboard computer(Ex move North at 10 m/s)

- **Onboard Computer, Sensors, & Middleware(Week 4, 5, 6)**
 - Jetson, Pi that run Ubuntu; this is the “brain” of our drone
 - This runs code (ROS FW - more formal implementation in Weeks 5/6) that tells our FC what to do
 - The code takes FC along with filtered sensor data (position, velocity, body rates, etc) as input.
 - Then, it processes the data using the logics/algorithms we implement, and generates fly commands.
 - Note: ROS helps us get data from and send commands to our hardware

2. What kind of challenges/projects can we work on as a Software team member?

A. We prioritize competitions, this revolves around various autonomy missions:

- Time-based waypoint flying
- Mapping, surveying

- Object detection, classification, and lastly localization
 - Depending on our approach, mapping could fit under this
- Having to drop packages/deliver payloads to targets

You can feel free to do an individual/group research project! (Note: this is a secondary focus)

Examples of interesting angles:

- GPS-denied navigation, mapping/localization
- (Learning-based) controller research
 - This is how we can make drones fly faster
- Swarming

B. Example Project:

- We use OpenDroneMap for image stitching
 - This isn't configured locally on the drone; UAV had to bring an extra operator last year which cooked them (it was also slower)
- The goal for this year is to do stitching in real-time, not all at once at the flight's end
 - This is challenging but will make a great project

3. What's the state of modern drone autonomy software?

A. Basic Autonomy vs. Production Autonomy

Feature	Basic Stack (Beginning Point)	Production Stack (What to work towards)
Flight control	Ardupilot/XP4/Beta flight	PX4 or ArduPilot on FC (with needed customization)
Higher-level software	Python scripts, basic onboard computer apps, or none	ROS 2 or personalized middleware on companion computers
Autonomy	Waypoint following	Perception + object localization + mission logic
Robustness/Safety	Simple failsafes	Redundant sensing, health monitoring, extensive logging, and more

Note: the industry is moving towards BVLOS (Beyond Visual Line of Sight) and multiple drones. By multiple drones we mean, multiple drones via one operator working in combination (Swarming)!

4. How does one learn about drones effectively in 2026?

A. AI

- It can help a ton!
 - It is super great for conceptual questions e.g, “what is SLAM” or “How do drones stay balanced”
 - This is great for help with code if used correctly and even greater with human expertise (Note: you will still have to proof-read this!)
 - This is amazing for explaining papers!
- **EXPERIENCE EXPERIENCE EXPERIENCE**
 - That’s what you come here for!

B. A Good workflow for learning new subjects:

1. Talk back and forth with AI, have it quiz you to reinforce that you understand things conceptually
2. Set goals for yourself! DO PROJECTS! This is the purpose of skill boosters!
3. Come up with methods to track your progress so you have motivation and feel like you are moving forward
4. With individual details, a combination of AI and fact-checking on individual sites/experience is amazing. AI doesn’t do great with specifics.
5. If you need help, reach out to others or the leaders!

Note: Also don’t be afraid to learn from different subteams/get involved in physical parts of the drone.